

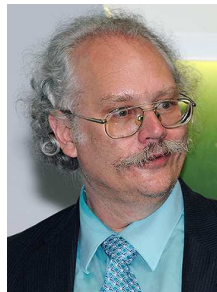
Some Notes on Post-Quantum Cryptography



J. Maurice Rojas

Texas A&M University

March 27, 2014



Flight

- **≤ 150 B.C.E.:**
Reference to
Śyena=Garuda
=a Bird-like
Hindu divinity
in India



Flight

- ≤ 150 B.C.E.: ...Garuda in India
- ≤ 100 B.C.E.: Greek legend of Icarus



Flight

- **≤ 150 B.C.E.:**
...Garuda in India
- **≤ 100 B.C.E.:**
Greek legend of
Icarus
- **559:** Manned kite
glide off a tower in
China



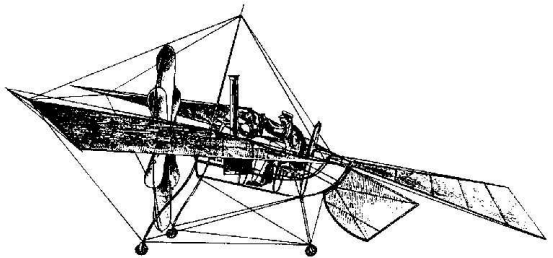
Flight

- **≤ 150 B.C.E.:**
...Garuda in India
- **≤ 100 B.C.E.:**
Greek legend of Icarus
- **559:** Manned kite glide off a tower in China
- **March 10, 1912:**
First bombing mission, during Italo-Turkish War



Flight

- **≤ 150 B.C.E.:**
...Garuda in India
- **≤ 100 B.C.E.:**
Greek legend of Icarus...
- **1874:** Félix du Temple makes short flight in steam-powered aluminum monoplane in France



Flight

- **≤ 150 B.C.E.:**
...Garuda in India
- **≤ 100 B.C.E.:**
Greek legend of Icarus...
- **1874:** Félix du Temple makes short flight in steam-powered aluminum monoplane in France
- **1914-1918:**
WWI



Flight

- **≤ 150 B.C.E.:**
...Garuda in India
- **≤ 100 B.C.E.:**
Greek legend of
Icarus...
- **1874:** Félix du
Temple makes
short flight...
- **1914-1918:**
WWI
- **1920:** National
Air Mail begins



Historical Context...

Flight

- **≤ 150 B.C.E.:**
...Garuda in India
- **≤ 100 B.C.E.:**
Greek legend of Icarus...
- **1874:** Félix du Temple makes short flight...
- **1914-1918:**
WWI
- **1920:** National Air Mail begins

Nuclear Energy

- **1789:** Martin Klaproth discovers uranium



Historical Context...

Flight

- **≤ 150 B.C.E.:**
...Garuda in India
- **≤ 100 B.C.E.:**
Greek legend of Icarus...
- **1874:** Félix du Temple makes short flight...
- **1914-1918:**
WWI
- **1920:** National Air Mail begins

Nuclear Energy

- **1789:** Martin Klaproth discovers uranium
- **1905:** Einstein's paper putting forth $\text{mass}=\text{energy}$



Historical Context...

Flight

- **≤ 150 B.C.E.:**
...Garuda in India
- **≤ 100 B.C.E.:**
Greek legend of Icarus...
- **1874:** Félix du Temple makes short flight...
- **1914-1918:**
WWI
- **1920:** National Air Mail begins

Nuclear Energy

- **1789:** Martin Klaproth discovers uranium
- **1905:** Einstein's paper putting forth $\text{mass}=\text{energy}$
- **1939:** Otto Frisch confirms energy release from fission



Historical Context...

Flight

- **≤ 150 B.C.E.:**
...Garuda in India
- **≤ 100 B.C.E.:**
Greek legend of Icarus...
- **1874:** Félix du Temple makes short flight...
- **1914-1918:**
WWI
- **1920:** National Air Mail begins

Nuclear Energy

- **1789:** Martin Klaproth discovers uranium
- **1905:** Einstein's paper putting forth $\text{mass}=\text{energy}$
- **1939:** Otto Frisch confirms energy release from fission
- **July 16, 1945:** First fission explosion



Historical Context...

Flight

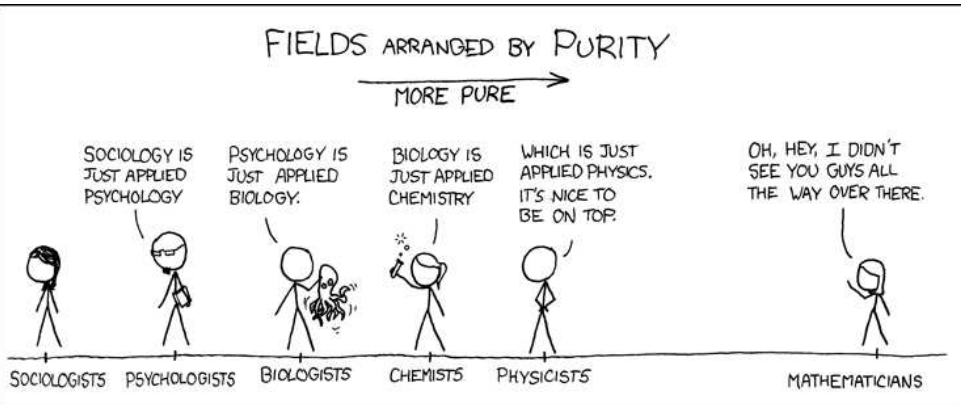
- **≤ 150 B.C.E.:**
...Garuda in India
- **≤ 100 B.C.E.:**
Greek legend of Icarus...
- **1874:** Félix du Temple makes short flight...
- **1914-1918:**
WWI
- **1920:** National Air Mail begins

Nuclear Energy

- **1789:** Martin Klaproth discovers uranium
- **1905:** Einstein's paper putting forth $\text{mass}=\text{energy}$
- **1939:** Otto Frisch confirms energy release from fission
- **July 16, 1945:** First fission explosion
- **December 1951:** First nuclear reactor, by Argonne National Labs, in Idaho



On to Crypto...



Historical Context...

Flight

- **≤ 150 B.C.E.:** Garuda...
- **≤ 100 B.C.E.:** Icarus...
- **1874:** du Temple's flight...
- **1914-1918:** WWI
- **1920:** Air Mail begins

Nuclear Energy

- **1789:** uranium
- **1905:** Einstein's paper...
- **1939:** Frisch confirms fission...
- **1945:** Fission explosion
- **1951:** First reactor

Cryptology

- **≤ 1500 B.C.E.:** Encrypted recipe on clay tablets in Mesopotamia



Historical Context...

Flight

- **≤ 150 B.C.E.:**
Garuda...
- **≤ 100 B.C.E.:**
Icarus...
- **1874:** du Temple's flight...
- **1914-1918:**
WWI
- **1920:** Air Mail begins

Nuclear Energy

- **1789:**
uranium
- **1905:**
Einstein's paper...
- **1939:** Frisch confirms fission...
- **1945:** Fission explosion
- **1951:** First reactor

Cryptology

- **≤ 1500 B.C.E.:**
Encrypted recipe on clay tablets in Mesopotamia
- **$\leq 1865:$**
Ciphers in US Civil War



Historical Context...

Flight

- **≤ 150 B.C.E.:**
Garuda...
- **≤ 100 B.C.E.:**
Icarus...
- **1874:** du
Temple's
flight...
- **1914-1918:**
WWI
- **1920:** Air
Mail begins

Nuclear Energy

- **1789:**
uranium
- **1905:**
Einstein's
paper...
- **1939:** Frisch
confirms
fission...
- **1945:** Fission
explosion
- **1951:** First
reactor

Cryptology

- **≤ 1500
B.C.E.:**
Encrypted
recipe on clay
tablets in
Mesopotamia
- **$\leq 1865:$**
Ciphers in US
Civil War
- **1920 – 1945:**
Enigma
invented by
Germans then
used in WWII



Historical Context...

Flight

- **≤ 150 B.C.E.:**
Garuda...
- **≤ 100 B.C.E.:**
Icarus...
- **1874:** du
Temple's
flight...
- **1914-1918:**
WWI
- **1920:** Air
Mail begins

Nuclear Energy

- **1789:**
uranium
- **1905:**
Einstein's
paper...
- **1939:** Frisch
confirms
fission...
- **1945:** Fission
explosion
- **1951:** First
reactor

Cryptology

- **≤ 1500
B.C.E.:**
Mesopotamia...
- **≤ 1865 :**
Ciphers in US
Civil War
- **1920 – 1945:**
Enigma
invented by
Germans then
used in WWII
- **1976 & 1978:**
DH and RSA
invented



Historical Context...

Flight

- **≤ 150 B.C.E.:** Garuda...
- **≤ 100 B.C.E.:** Icarus...
- **1874:** du Temple's flight...
- **1914–1918:** WWI
- **1920:** Air Mail begins

Nuclear Energy

- **1789:** uranium
- **1905:** Einstein's paper...
- **1939:** Frisch confirms fission...
- **1945:** Fission explosion
- **1951:** First reactor

Cryptology

- **$\dots \leq 1945 \dots$**
- **≤ 1973 –'74:** RSA and DH actually invented at GCHQ!



Historical Context...

Flight

- **≤ 150 B.C.E.:** Garuda...
- **≤ 100 B.C.E.:** Icarus...
- **1874:** du Temple's flight...
- **1914–1918:** WWI
- **1920:** Air Mail begins

Nuclear Energy

- **1789:** uranium
- **1905:** Einstein's paper...
- **1939:** Frisch confirms fission...
- **1945:** Fission explosion
- **1951:** First reactor

Cryptology

- **$\dots \leq 1945 \dots$**
- **≤ 1973 –'74:** RSA and DH **actually** invented at GCHQ!
- **1994:** Peter Shor finds quantum algorithm that, in theory, easily breaks DH and RSA.



Historical Context...

Flight

- **≤ 150 B.C.E.:** Garuda...
- **≤ 100 B.C.E.:** Icarus...
- **1874:** du Temple's flight...
- **1914–1918:** WWI
- **1920:** Air Mail begins

Nuclear Energy

- **1789:** uranium
- **1905:** Einstein's paper...
- **1939:** Frisch confirms fission...
- **1945:** Fission explosion
- **1951:** First reactor

Cryptology

- **$\dots \leq 1945 \dots$**
- **≤ 1973 –'74:** RSA and DH **actually** invented at GCHQ!
- **1994:** Peter Shor finds quantum algorithm that, in theory, easily breaks DH and RSA.
- **1998:** GCHQ: “Yeah, we did invent RSA and DH...”



What are DH and RSA? Who Cares?

- 1 DH and RSA are methods for sharing encryption keys, now used in routers and internet servers **everywhere**...



What are DH and RSA? Who Cares?

- 1 DH and RSA are methods for sharing encryption keys, now used in routers and internet servers **everywhere**...
- 2 DH is based on the *Discrete Log Problem*: Given $a, g, p \in \mathbb{N}$, find $x \in \{0, \dots, p-1\}$ with $g^x = a \bmod p$.



What are DH and RSA? Who Cares?

- 1 DH and RSA are methods for sharing encryption keys, now used in routers and internet servers **everywhere**...
- 2 DH is based on the *Discrete Log Problem*: Given $a, g, p \in \mathbb{N}$, find $x \in \{0, \dots, p-1\}$ with $g^x = a \bmod p$.
- 3 RSA is based on the “hardness” of *Integer Factoring*:



What are DH and RSA? Who Cares?

- 1 DH and RSA are methods for sharing encryption keys, now used in routers and internet servers **everywhere**...
- 2 DH is based on the *Discrete Log Problem*: Given $a, g, p \in \mathbb{N}$, find $x \in \{0, \dots, p-1\}$ with $g^x = a \bmod p$.
- 3 RSA is based on the “hardness” of *Integer Factoring*: Given that n is a product of two distinct primes, find the primes!



What are DH and RSA? Who Cares?

- ➊ DH and RSA are methods for sharing encryption keys, now used in routers and internet servers **everywhere**...
- ➋ DH is based on the *Discrete Log Problem*: Given $a, g, p \in \mathbb{N}$, find $x \in \{0, \dots, p-1\}$ with $g^x = a \bmod p$.
- ➌ RSA is based on the “hardness” of *Integer Factoring*: Given that n is a product of two distinct primes, find the primes!
- ➍ As far as we know (on the outside), there are no *practical* quantum computers... Unless you count D-Wave’s 128-“qubit” computer which costs \$10M...



What are DH and RSA? Who Cares?

- 1 DH and RSA are methods for sharing encryption keys, now used in routers and internet servers **everywhere**...
- 2 DH is based on the *Discrete Log Problem*: Given $a, g, p \in \mathbb{N}$, find $x \in \{0, \dots, p-1\}$ with $g^x = a \bmod p$.
- 3 RSA is based on the “hardness” of *Integer Factoring*: Given that n is a product of two distinct primes, find the primes!
- 4 As far as we know (on the outside), there are no *practical* quantum computers... Unless you count D-Wave’s 128-“qubit” computer which costs \$10M...
- 5 So what?



Why Cares if Quantum Computing Comes?

As we've already seen from the 25 year lag of “inside” discovery disclosure (and the some recent controversial leaks), “inside” completion of a quantum computer is likely to remain hidden, be it by NSA



Why Cares if Quantum Computing Comes?

As we've already seen from the 25 year lag of “inside” discovery disclosure (and the some recent controversial leaks), “inside” completion of a quantum computer is likely to remain hidden, be it by NSA or Google



Why Cares if Quantum Computing Comes?

As we've already seen from the 25 year lag of “inside” discovery disclosure (and the some recent controversial leaks), “inside” completion of a quantum computer is likely to remain hidden, be it by NSA or Google or IBM



Why Cares if Quantum Computing Comes?

As we've already seen from the 25 year lag of “inside” discovery disclosure (and the some recent controversial leaks), “inside” completion of a quantum computer is likely to remain hidden, be it by NSA or Google or IBM or Siemens



Why Cares if Quantum Computing Comes?

As we've already seen from the 25 year lag of “inside” discovery disclosure (and the some recent controversial leaks), “inside” completion of a quantum computer is likely to remain hidden, be it by NSA or Google or IBM or Siemens or Sony



Why Cares if Quantum Computing Comes?

As we've already seen from the 25 year lag of “inside” discovery disclosure (and the some recent controversial leaks), “inside” completion of a quantum computer is likely to remain hidden, be it by NSA or Google or IBM or Siemens or Sony or Dubai...



Why Cares if Quantum Computing Comes?

As we've already seen from the 25 year lag of “inside” discovery disclosure (and the some recent controversial leaks), “inside” completion of a quantum computer is likely to remain hidden, be it by NSA or Google or IBM or Siemens or Sony or Dubai...

“OK, so I'll stop using public-key crypto now...”



Why Cares if Quantum Computing Comes?

As we've already seen from the 25 year lag of “inside” discovery disclosure (and the some recent controversial leaks), “inside” completion of a quantum computer is likely to remain hidden, be it by NSA or Google or IBM or Siemens or Sony or Dubai...

“OK, so I'll stop using public-key crypto now...”

Not so fast...



Why Cares if Quantum Computing Comes?

As we've already seen from the 25 year lag of “inside” discovery disclosure (and the some recent controversial leaks), “inside” completion of a quantum computer is likely to remain hidden, be it by NSA or Google or IBM or Siemens or Sony or Dubai...

“OK, so I'll stop using public-key crypto now...”

Not so fast... What about archival data?



Why Cares if Quantum Computing Comes?

As we've already seen from the 25 year lag of “inside” discovery disclosure (and the some recent controversial leaks), “inside” completion of a quantum computer is likely to remain hidden, be it by NSA or Google or IBM or Siemens or Sony or Dubai...

“OK, so I'll stop using public-key crypto now...”

Not so fast... What about archival data? e.g., medical records



Why Cares if Quantum Computing Comes?

As we've already seen from the 25 year lag of “inside” discovery disclosure (and the some recent controversial leaks), “inside” completion of a quantum computer is likely to remain hidden, be it by NSA or Google or IBM or Siemens or Sony or Dubai...

“OK, so I'll stop using public-key crypto now...”

Not so fast... What about archival data? e.g., medical records, contracts, etc...



Why Cares if Quantum Computing Comes?

As we've already seen from the 25 year lag of “inside” discovery disclosure (and the some recent controversial leaks), “inside” completion of a quantum computer is likely to remain hidden, be it by NSA or Google or IBM or Siemens or Sony or Dubai...

“OK, so I'll stop using public-key crypto now...”

Not so fast... What about archival data? e.g., medical records, contracts, etc...

Experts say we *could* have an honest, working quantum computer within 15 years...



Complexity Theory vs. Practical Complexity

Definition

We say two functions $f, g : \mathbb{N} \longrightarrow \mathbb{R}$ satisfy $f = O(g) \iff$ there are constants $C, M > 0$ with $|f(x)| \leq M g(x)$ for all $x \geq C$. $\diamond$



Complexity Theory vs. Practical Complexity

Definition

We say two functions $f, g : \mathbb{N} \longrightarrow \mathbb{R}$ satisfy $f = O(g) \iff$ there are constants $C, M > 0$ with $|f(x)| \leq M g(x)$ for all $x \geq C$. $\diamond$

Examples



Complexity Theory vs. Practical Complexity

Definition

We say two functions $f, g : \mathbb{N} \rightarrow \mathbb{R}$ satisfy $f = O(g) \iff$ there are constants $C, M > 0$ with $|f(x)| \leq M g(x)$ for all $x \geq C$. $\diamond$

Examples

- Inverting an $n \times n$ matrix takes $O(n^3)$ arithmetic operations (or $O(n^{2.3727})$ if you're *really* clever).



Complexity Theory vs. Practical Complexity

Definition

We say two functions $f, g : \mathbb{N} \rightarrow \mathbb{R}$ satisfy $f = O(g) \iff$ there are constants $C, M > 0$ with $|f(x)| \leq M g(x)$ for all $x \geq C$. $\diamond$

Examples

- Inverting an $n \times n$ matrix takes $O(n^3)$ arithmetic operations (or $O(n^{2.3727})$ if you're *really* clever).
- Factoring an N -digit integer takes $2^{O(N^{1/3}(\lg N)^{2/3})}$ seconds on our best (classical) computers.



Complexity Theory vs. Practical Complexity

Definition

We say two functions $f, g : \mathbb{N} \rightarrow \mathbb{R}$ satisfy $f = O(g) \iff$ there are constants $C, M > 0$ with $|f(x)| \leq M g(x)$ for all $x \geq C$. $\diamond$

Examples

- Inverting an $n \times n$ matrix takes $O(n^3)$ arithmetic operations (or $O(n^{2.3727})$ if you're *really* clever).
- Factoring an N -digit integer takes $2^{O(N^{1/3}(\lg N)^{2/3})}$ seconds on our best (classical) computers.

But what about $N = 200$, say?



Complexity Theory vs. Practical Complexity

Definition

We say two functions $f, g : \mathbb{N} \rightarrow \mathbb{R}$ satisfy $f = O(g) \iff$ there are constants $C, M > 0$ with $|f(x)| \leq M g(x)$ for all $x \geq C$. $\diamond$

Examples

- Inverting an $n \times n$ matrix takes $O(n^3)$ arithmetic operations (or $O(n^{2.3727})$ if you're *really* clever).
- Factoring an N -digit integer takes $2^{O(N^{1/3}(\lg N)^{2/3})}$ seconds on our best (classical) computers.

But what about $N=200$, say? Or $N=500$?



Complexity Theory vs. Practical Complexity

Definition

We say two functions $f, g : \mathbb{N} \longrightarrow \mathbb{R}$ satisfy $f = o(g) \iff \lim_{x \rightarrow +\infty} \frac{f(x)}{g(x)} = 0$. $\diamond$



Complexity Theory vs. Practical Complexity

Definition

We say two functions $f, g : \mathbb{N} \longrightarrow \mathbb{R}$ satisfy $f = o(g) \iff \lim_{x \rightarrow +\infty} \frac{f(x)}{g(x)} = 0$. $\diamond$

Examples



Complexity Theory vs. Practical Complexity

Definition

We say two functions $f, g : \mathbb{N} \rightarrow \mathbb{R}$ satisfy $f = o(g) \iff \lim_{x \rightarrow +\infty} \frac{f(x)}{g(x)} = 0$. $\diamond$

Examples

- Shor's Algorithm can factor N -digit RSA integers within $N^{2+o(1)}$ seconds, on a quantum computer with $N^{1+o(1)}$ qubits.



Complexity Theory vs. Practical Complexity

Definition

We say two functions $f, g : \mathbb{N} \rightarrow \mathbb{R}$ satisfy $f = o(g) \iff \lim_{x \rightarrow +\infty} \frac{f(x)}{g(x)} = 0$. $\diamond$

Examples

- Shor's Algorithm can factor N -digit RSA integers within $N^{2+o(1)}$ seconds, on a quantum computer with $N^{1+o(1)}$ qubits.
- A 1978 cryptosystem of McEliece



Complexity Theory vs. Practical Complexity

Definition

We say two functions $f, g : \mathbb{N} \rightarrow \mathbb{R}$ satisfy $f = o(g) \iff \lim_{x \rightarrow +\infty} \frac{f(x)}{g(x)} = 0$. $\diamond$

Examples

- Shor's Algorithm can factor N -digit RSA integers within $N^{2+o(1)}$ seconds, on a quantum computer with $N^{1+o(1)}$ qubits.
- A 1978 cryptosystem of McEliece (quantum-resistant so far!)



Complexity Theory vs. Practical Complexity

Definition

We say two functions $f, g : \mathbb{N} \rightarrow \mathbb{R}$ satisfy $f = o(g) \iff \lim_{x \rightarrow +\infty} \frac{f(x)}{g(x)} = 0$. $\diamond$

Examples

- Shor's Algorithm can factor N -digit RSA integers within $N^{2+o(1)}$ seconds, on a quantum computer with $N^{1+o(1)}$ qubits.
- A 1978 cryptosystem of McEliece (quantum-resistant so far!) needs a key-size of $b^{2+o(1)}$ to force breakage to take $\geq 2^b$ seconds.



Complexity Theory vs. Practical Complexity

Definition

We say two functions $f, g : \mathbb{N} \rightarrow \mathbb{R}$ satisfy $f = o(g) \iff \lim_{x \rightarrow +\infty} \frac{f(x)}{g(x)} = 0$. $\diamond$

Examples

- Shor's Algorithm can factor N -digit RSA integers within $N^{2+o(1)}$ seconds, on a quantum computer with $N^{1+o(1)}$ qubits.
- A 1978 cryptosystem of McEliece (quantum-resistant so far!) needs a key-size of $b^{2+o(1)}$ to force breakage to take $\geq 2^b$ seconds.
- RSA needs a key-size of $0.016...b^3/(\lg b)^2$ to force breakage to take $\geq 2^b$ seconds (on a classical computer).



Complexity Theory vs. Practical Complexity

Definition

We say two functions $f, g : \mathbb{N} \rightarrow \mathbb{R}$ satisfy $f = o(g) \iff \lim_{x \rightarrow +\infty} \frac{f(x)}{g(x)} = 0$. $\diamond$

Examples

- Shor's Algorithm can factor N -digit RSA integers within $N^{2+o(1)}$ seconds, on a quantum computer with $N^{1+o(1)}$ qubits.
- A 1978 cryptosystem of McEliece (quantum-resistant so far!) needs a key-size of $b^{2+o(1)}$ to force breakage to take $\geq 2^b$ seconds.
- RSA needs a key-size of $0.016...b^3/(\lg b)^2$ to force breakage to take $\geq 2^b$ seconds (on a classical computer).

But the devil is in the constants!: real-world security needs around $b=128 \implies$ thousands of bits for RSA but millions of bits for McEliece [Bernstein, 2009]!



Moral: Deep Understanding Needed!

- If we are to maintain our comfortable lives (with reasonable privacy), we need to get work **now** on improving alternative cryptosystems!



Moral: Deep Understanding Needed!

- If we are to maintain our comfortable lives (with reasonable privacy), we need to get work **now** on improving alternative cryptosystems!
- A particularly promising vein of quantum-resistant cryptosystems come from *lattice vector problems*: these cryptosystems are closely related to **NP**-hard problems that have good average-case behavior: *average instances are provably almost as hard as worst-case instances*.



Moral: Deep Understanding Needed!

- If we are to maintain our comfortable lives (with reasonable privacy), we need to get work **now** on improving alternative cryptosystems!
- A particularly promising vein of quantum-resistant cryptosystems come from *lattice vector problems*: these cryptosystems are closely related to **NP**-hard problems that have good average-case behavior: *average instances are provably almost as hard as worst-case instances*.

Example: The *Closest Vector Problem (CVP)* is: Given vectors $v_1, \dots, v_N \in \mathbb{Z}^n$ and a target vector $t \in \mathbb{Q}^n$, find the closest *integer* linear combination $\alpha_1 v_1 + \dots + \alpha_n v_n$ to t .



Moral: Deep Understanding Needed!

- If we are to maintain our comfortable lives (with reasonable privacy), we need to get work **now** on improving alternative cryptosystems!
- A particularly promising vein of quantum-resistant cryptosystems come from *lattice vector problems*: these cryptosystems are closely related to **NP**-hard problems that have good average-case behavior: *average instances are provably almost as hard as worst-case instances*.

Example: The *Closest Vector Problem (CVP)* is: Given vectors $v_1, \dots, v_N \in \mathbb{Z}^n$ and a target vector $t \in \mathbb{Q}^n$, find the closest *integer* linear combination $\alpha_1 v_1 + \dots + \alpha_n v_n$ to t .

- Solving CVP, even within a factor of $\sqrt{n}$, in time polynomial in n would imply **P=NP**, i.e., you would simultaneously solve numerous other important problems in polynomial-time (and win \$1M)!





Thank you for your attention.

See:

- Bernstein, Buchmann, Dahmen's 2009 Springer book on Post-Quantum Cryptography
- Web-sites of Dan Bernstein and Tanja Lange
- Blog of Scott Aaronson
- Math 470
- Math 415
- Math 427...

...and www.math.tamu.edu/~rojas for further info on algorithmic algebraic geometry.

